Envoy RTL: Ubuntu CI compatibility

2026-10-18 Run batch: `envoy-fpga-20261018-ci-compat`

The PR's module jobs passed core tests, then failed RTL lint under Ubuntu's Verilator 5.020. Earlier local validation used 5.032. The unchanged parent reproduces seven array-loop errors and four fatal width warnings with 5.020. The fix raises the static loop-unrolling limit to 256 and explicitly zero-extends each 23-bit NTT multiplier operand to 32 bits. Fatal `-Wall`, assertions, test cases and structural checks remain enabled.

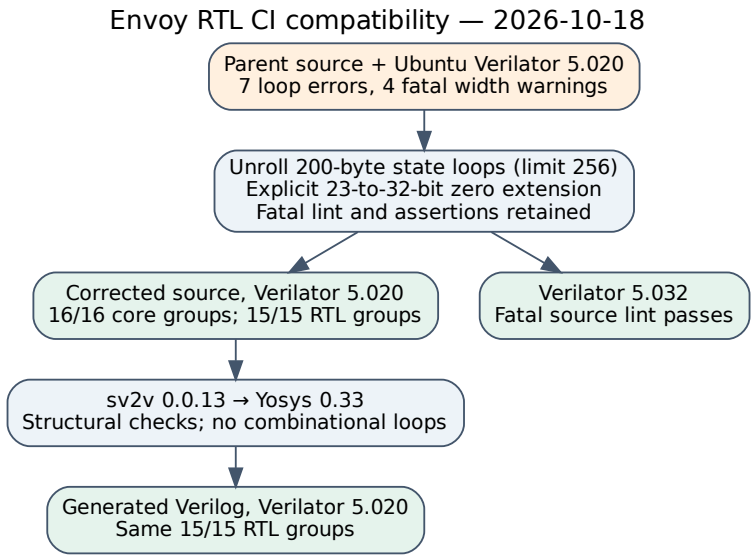


Figure 1. Validation stages, not performance measurements. The synthesis result repeats the same test groups; it is not a formal equivalence proof or 30 independent specifications. Source: `flow.dot`.

Check	Observed result
Core transport	16/16 groups passed.
Source RTL, 5.020	Fatal lint passed; 15/15 groups passed.
Yosys 0.33 replay	Structural checks passed; same 15/15 groups passed.
Source lint, 5.032	Passed; full simulation was not repeated with this version.
Scoped workflow	actionlint 1.7.7 passed.
Root Rust gate	Exit 101 before tests: same 26 errors in unchanged source.

Environment and limits. Debian 13, GCC 14.2.0, OpenSSL 3.5.7; Ubuntu packages Verilator 5.020 and Yosys 0.33; sv2v 0.0.13. The RTL tool versions match CI, but the full hosted runner environment does not. Fixtures and independent oracles are unchanged; seed `0x4650433252544c`. No optional liboqs/BoringSSL/live-Envoy suite, physical FPGA test or cryptographic timing measurement was run.

Retained failures and gates. Both failing module CI logs, the local reproduction, an unroll-only attempt, package-relocation failures and an actionlint working-directory error are retained locally. Existing workflow duplicate keys, curve-registry/name failures and root Rust errors remain outside this change. No submitted reviews or inline threads were present in the initial snapshot; neutral Cursor checks reported usage limits. New-head CI and actual review requirements must still be satisfied before merging.

Provenance. Extends parent `e6369b99328ab2b6ee301679c133691000c1f31a`, PR #2122. The summary receipt records source hashes and aggregate outcomes. The new local raw archive and frozen run manifest await upload approval; their hashes identify the execution-era snapshot. The full report records scope and reproduction. Older evidence remains unchanged. Complete crypto controllers, entropy/key custody, FPC1/F2/AFI integration, physical and side-channel validation, and measured acceleration remain open.